



Precision cancer medicine: What has translated into clinical use in Belgium?

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ARTICLE INFO

Keywords:

Precision oncology
Next-generation sequencing
Knowledge translation
Genomics
Knowledge transfer
Implementation assessment

ABSTRACT

Rationale: In 2016, Belgium launched the Next Generation Sequencing (NGS) Roadbook, consisting in 10 Actions, across the health care system, to facilitate the uptake of NGS in routine clinical practice. We compiled feedback on deployment of the NGS Roadbook from governmental stakeholders and beneficiaries: health policy makers, insurance providers, pathologists, geneticists, and oncologists.

Main findings: The Roadbook ensured the establishment of a Commission on Personalized Medicine and NGS testing guidelines. A national benchmarking trial ensued, and 10 networks of hospitals and laboratories adopted a reimbursement convention with the Belgian Health Insurance Agency. The Healthdata.be platform centralizes the collection of NGS metrics, and citizens were consulted on their views about NGS and genomics.

Conclusion: The Roadbook facilitated the implementation of NGS in routine (hemato-)oncology care in Belgium. Some challenges remain linked to data sharing and access by a wider range of stakeholders. Next steps include continuous monitoring of health outcomes and the budgetary impact.

1. Introduction

Cancer is one of the main public health challenges in Europe and the second leading cause of mortality in EU countries. While over 40 % of cases are preventable [1], still 3.9 million people in the EU are diagnosed each year, and 1.9 million deaths are cancer-related [2]. At EU level, there have been considerable investments in developing cancer control guidelines and recommendations, building on the outcomes of joint efforts between the Member States and the European Commission (Joint Actions EPACC, CanCON). Now, the time has come for implementation and a better understanding of how quality improvements and innovations have been applied.

Precision oncology is currently at the heart of innovations in cancer research and care [3]. Among which, next-generation sequencing (NGS) facilitates the targeting of causative mutations for tumours [4,5]. However, there can be significant challenges in leveraging this technology in practice. For genomic testing, high-quality tissue needs to be sampled and analyzed in laboratories equipped to perform NGS. Next, clinicians need to be familiar with the latest testing and resampling

methodologies, and the types of genomic variants reported by laboratories. Data should also be stored according to EU and national data privacy frameworks [6]. One difficulty is the nature and volume of data which need to be integrated for decision-making, ranging from -omics, imaging, lifestyle and environmental determinants [7]. Finally, multi-disciplinary input from (hemato-)oncologists and allied health professionals is required to report and communicate results in a patient-centered approach.

In 2015, the Belgian Cancer Centre and the Healthcare Knowledge Centre (KCE) conducted a state of play of NGS use in (hemato-)oncology in Belgium (Belgian Health Care Knowledge Center Report 240) [6]. From this study, emerged the need to harmonize NGS testing practices in Belgium and facilitate the uptake in clinical settings. More specifically, hospitals and laboratories were not using the same tests and nomenclature which impacted reimbursements by the National Institute for Health and Disability Insurance (INAMI/RIZIV), and access for targeted therapies. The recommendations from this feasibility study were officially approved by the Ministry of Social Affairs and Health in 2016, and endorsed by the main stakeholders involved in oncology care and

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service delivery in Belgium [8]. The stakeholders endorsed 10 inter-connected Actions to be deployed from 2016 to 2021 (listed in Table 1) which underpin the Belgian NGS Roadbook. The Roadbook aimed to guide the uptake of NGS in routine clinical practice in Belgium by harmonizing NGS testing practices across clinical settings. Of note, to monitor the uptake of NGS testing in (hemato-)oncology in clinical practice, a reimbursement convention between laboratories performing (hemato-)oncology testing and the National Institute for Health and Disability Insurance (INAMI/RIZIV, <https://www.riziv.fgov.be>), was set up as a pilot phase for the period 2019–2022. At this stage, it is important to get a clear overview of how the the NGS Roadbook was rolled out in practice: what went well, what could be improved and what the outlooks could be. Therefore, this study aimed to evaluate: i) whether the NGS Roadbook was deployed in the Belgian health care system as planned, ii) and what impact the NGS Roadbook had on stakeholders' capacity to use NGS in (hemato-)oncology.

1.1. Overview of the NGS Roadbook intervention

As shown in Table 1, the NGS Roadbook consisted in a comprehensive set of actions targeting care providers, laboratories, the health insurance system, citizens, and health policy makers. We have listed these in Table 1 and we provide a timeline for when the main objectives were reached. The Roadbook outlined the provision of a governance structure for the intervention (Action 1), technical and logistical requirements of NGS testing and use (Action 2–5), organizational specifications (Action 6,10), the allocated budget and incentives for participating laboratories (Action 9), and targets for capacity-building and raising awareness among health professionals, patients and citizens (Action 7,8). A total budget of 5,122,000 Euros was made available for this; the budget calculations were based on the KCE feasibility study in which NGS testing costs were estimated at 350€ for equipment including services, reagents and personnel [6]. The all inclusive cost for a NGS panel test limited to DNA alterations of direct clinical utility varies by platform, and the maximum number of NGS panel tests per year for Belgium is 7000–10,000.

1.2. Conceptual framework for the implementation assessment

In general, implementation assessments are conducted to clarify what direction an intervention has taken based on priorities, resources, time, and skills. These assessments help construct a shared understanding with stakeholders about the purpose and value-chain of the intervention. Implementation assessments foster transparency, and are essential in providing benchmarks and pointing out what corrective actions need to be taken. While there are many frameworks that could be used to evaluate health interventions, in 2018, a Health-Information (HI)-Impact framework was developed specifically to monitor the uptake of national health information products (e.g. guidelines and recommendations) in policy and practice [9]. The HI-Impact framework addresses the determinants of the use of evidence for decision-making and health intervention, also known as knowledge translation. The four pillars of the HI-Impact framework are described below:

- **Health Information and Evidence Quality:** monitors the reliability, accuracy and relevance of the evidence provided at national/regional level.
- **Health Information System Responsiveness:** monitors the availability and access of the evidence – including dissemination efforts targeting a wide range of stakeholders: policy makers, researchers, health care professionals and the general public.
- **Stakeholder Engagement:** monitors the use of the evidence in the health care system (i.e. for capacity-building, decision-making and intervention).

Table 1

Description of NGS Roadbook Actions.

Action title	Coordinator	Main deliverable and dates of achievement
ACTION 1: Establish the Commission on Personalized Medicine (ComPerMed)	Sciensano, Cancer Center	The Belgian Commission of Personalized Medicine (ComPerMed) was created in 2015 to act a national advisory board on new developments and insights in personalized medicine. The list of current working group members is available from the website (https://www.compermed.be/en/org/organisation). ComPerMed advises the Belgian Health Insurance Agency about the practices and the tests to be reimbursed.
ACTION 2: Development of guidelines for NGS testing and variant interpretation in (hemato-)oncology	ComPerMed	Guidelines on NGS testing were published in 2017 and updated in 2019 (BELAC: https://www.ariez.nl/wp-content/uploads/2018/08/56-67.pdf). The recommendations are complementary to the International Organization for Standardization (ISO) 15189 standard (medical laboratories) and to other international NGS guidelines.
ACTION 3: Development of standardized indications for NGS use in (hemato-)oncology	ComPerMed	Workflows are published annually and available from the ComPerMed website for the period 2018–2021 for diagnosis, prognosis and therapy, including the list of genes and gene-regions to be analyzed by NGS for specific cancers: solid tumours (N = 18); haemato (N = 14)
ACTION 4&5: Implementation of benchmarking trials in clinical settings/ laboratories, and launch of a national External Quality Assessment for NGS use in (hemato-)oncology	Quality of Laboratory Dpt., Sciensano	7 EQA benchmarks were conducted in 16–20 participating laboratories from 2017 to 2021. The EQA reports are available from https://www.wiv-isp.be/QML/ . The reports include the evaluation criteria and outcomes from each of the laboratories.
ACTION 6: Implementation of a technical platform for the central collection and storage of NGS data	Healthdata.be, Sciensano	NGS data are included in a uniform and secure manner within the eHealth "Healthdata.be" environment. The first NGS data registration was recorded in 2019
ACTION 7: Provide NGS education and training to oncologists, medical students and allied health professions	Belgian WG of Molecular Pathology	An annual training course was set up by the Belgian Working Group of Molecular Pathology (the course was offered in 2019 and 2020) The Certificate in Molecular pathology includes 6 courses https://belgian-society-pathology.eu/working-groups/molecular
ACTION 8: Engaging patients on informed consent, legal and ethical implications of NGS use in (hemato-)oncology	Sciensano, Cancer Center, Ethics unit	11 Focus groups were held in 2018–2019, 2 h sessions; DNA debate (debatadn.be) with 1250 contributions; citizen lab held (N = 32 participants)

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Table 1 (continued)

Action title	Coordinator	Main deliverable and dates of achievement
ACTION 9: Pilot study ‘NGS use in routine diagnostics’.	Sciensano, Cancer Center & RIZIV/INAMI (Sickness Fund)	Publishing guidelines and report are available from: https://www.e-cancer.be/site/s/default/files/rapport_focus_groups_fr.pdf The NGS convention is the reimbursement contract between the National Health and Disability Insurance fund and the signatory laboratories/testing centers. It was established in Feb 2019 and runs until 2022. The pilot study which relies on the convention allowed to implement NGS under controlled conditions and to monitor closely technical aspects, NGS data registration, and use of NGS guidelines by signatory laboratories and hospitals. Available from: https://www.inami.fgov.be/fr/professionnels/etablissements-services/laboratoires/Pages/oncologie-remboursement-biologie-moleculaire-ngs.aspx
ACTION 10: Build nodes of expertise: hospital networks for NGS use in (hemato)-oncology	Sciensano, Cancer Center RIZIV/INAMI	In 2019, ten networks were approved. These currently include 115 hospitals and 30 laboratories and the list is available from: https://www.inami.fgov.be/fr/professionnels/etablissements-services/laboratoires/Pages/oncologie-remboursement-biologie-moleculaire-ngs.aspx

NOTES: More detailed information on the implementation assessments for the NGS Roadbook actions is provided in Supplementary material.

- Knowledge Integration: assesses the use of evidence across sectors. This domain provides more insight on the sustainability of the evidence in the broader community setting, and on the societal impact.

We have used this framework to identify what worked or did not work in the deployment of the NGS Roadbook in Belgium, and to identify future developments.

1.3. Data collection

To conduct the assessment, we relied on qualitative data and content analysis. The coordinators of the 10 NGS Actions were invited to describe how they implemented the Roadbook by filling in the “NGS Action Evaluation sheet” (available in Supplementary material). They were asked about: the objectives of the Action, which performance measures were available, the resources mobilized, observed changes in the practice of target stakeholders, and the acceptability and fidelity of the intervention to the NGS Roadbook. This information was requested in each of the following stages of implementation: 1) exploration, 2) installation, 3) initial implementation, 4) full implementation, and 5) expansion and scale-up. Finally, the coordinators also flagged any modifications or delays to their intervention, including upcoming developments.

In order to cross-validate this information and deepen our case descriptions of the Actions, we also surveyed the beneficiaries of the NGS Roadbook (cf. survey in Supplementary Material). The ComPerMed Management Board, which gathers an expert group representing the

main actors of cancer surveillance, control and care in Belgium was contacted in June 2020. The questionnaire was developed in Limesurvey V2 and distributed online. There were 23 items in the survey based on our conceptual framework. Participants were given statements and asked to share their agreement or disagreement with the statements using a Likert scale (True-Somewhat true-Not true-NA). The statements aimed to capture the impact of the NGS Roadbook (content and format) with regards to quality, dissemination and use by stakeholders. Finally, we compiled the NGS Roadbook Action evaluation sheets and the ComPerMed survey results under the following subheadings which map onto the HI-Impact framework:

- 1 Defining a quality framework for NGS in (hemato)-oncology diagnostics in Belgium
- 2 Ensuring the dissemination of the NGS Quality framework
- 3 Stakeholder engagement and capacity building
- 4 Knowledge integration in civil society

2. Findings

We will highlight here the main markers of implementation and the challenges which needed to be overcome to implement the intervention. This includes commenting on the process as well as on the determinants of acceptability and evidence uptake by the target stakeholders. More information on the implemented NGS Actions is available as Supplementary material (cf. Action evaluation sheets 1–10). Possible next steps and recommendations for an extension of the Belgian NGS convention are provided in the discussion.

2.1. Defining a Quality framework for NGS in (hemato)-oncology diagnostics in Belgium

Quality frameworks are essential to set up a successful introduction of guidelines in practice. Quality components consist in ensuring that the underlying evidence is scientifically robust, but also that the appropriate governance and resources are allocated for uptake of the evidence in practice. In line with the NGS Roadbook, the Belgian Commission of Personalized Medicine (ComPerMed) was created in 2015 to act as a national advisory board on new developments and insights in personalized medicine. The list of current working group members (23 on the management board and 143 members in the Scientific Board) is available from the website (<https://www.compermed.be/en/organisation>). A charter established with RIZIV/INAMI (the National Institute for Health and Disability Insurance) provided the mandate. More specifically, ComPerMed informs coherent reimbursement decisions on companion diagnostics (Platform CDx) made at the RIZIV/INAMI. Within ComPerMed 3 different working groups have been set up: the solid tumor working group with 153 experts, the hematological tumors working group (62), and the NGS technology working group (98).

The management board currently includes experts representing the Ministry of Health, the Belgian Healthcare Knowledge Center KCE, the Belgian Health Insurance Agency (RIZIV/INAMI), the Federal Agency for Medicines and Health Products (FAGG/AFMPS), the Belgian Cancer Registry, the Belgian Society for Human Genetics, the College of Oncology, the Commission of Pathology and the Commission of Clinical Biology. From the ComPerMed management Board, 32 experts participated in the survey but only 14 filled in the survey evaluation completely on the impact of the NGS Roadbook. In Supplementary Table 2 we have presented the survey results and statements, and we have summarized the feedback by domain in Fig. 1. They nearly all agreed that the Roadbook approach facilitated the uptake of NGS technology in Belgium by providing a comprehensive overview of the need and actions required to introduce the technology in routine practice. We will refer to this table again throughout this section.

In terms of quality, another Roadbook objective was to set up

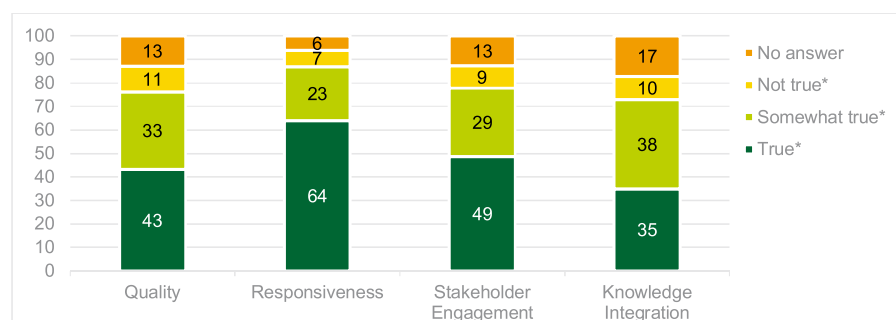


Fig. 1. ComPerMed feedback on the impact of the NGS Roadbook in policy and practice.

*Based on the % average: Quality (11 items), Responsiveness [3], Stakeholder engagement [5], Knowledge Integration [3].

“Quality” relates to monitoring NGS Roadbook production, accuracy, and content relevance; “Responsiveness” relates to monitoring NGS Roadbook access and availability; “Stakeholder Engagement” relates to monitoring the use of the NGS Roadbook in the healthcare system; “Knowledge Integration” relates to monitoring the use of the NGS Roadbook in civil society and across other policy and practice sectors.

normative guidelines for NGS use and variant interpretation in Belgium (Action 2,3). This was successfully implemented (see Table 1). The ComPerMed experts conducted a systematic evaluation of all NGS tests performed nationally. Different technologies were being used focused mainly on targeted DNA sequencing (small panels) and increasingly, RNA sequencing and larger panels are now more widely used. ComPerMed started with a systematic evaluation of all molecular tests currently performed in routine in Belgium for each tumour type (for 15 solid tumors and 10 hematological malignancies). These were annotated based on their utility (diagnosis, prognosis or therapy). The OncoKB test levels, a precision oncology knowledge base with information about the effects and treatment implications of specific cancer gene alterations [10], was adapted based on clinical guidelines and standard of care, expert’s opinion and approved drugs in Belgium. For each tumor type, the testing procedures were schematized and approved by consensus during expert deliberations. The validated workflows (<https://www.compermed.be/activites/workflows#/>) provide a technical framework for laboratories, that BELAC-auditors (National Accreditation Agency) and other experts can use. From a technical point of view, laboratories can choose the technology they want to sequence, but for the bioinformatics part, the NGS guidelines developed as part of the NGS Roadbook actions, facilitate and harmonise implementation, verification and validation of targeted NGS tests to detect acquired somatic mutations in DNA (or RNA), or acquired fusion genes in RNA, in (haemato)-oncology [11]. Initially, the guidelines were deemed too generic, but given the results of the benchmarking trials (see Actions 4/5), more detailed guidelines ensued that were variant-based and linked to the patient’s clinical report.

Guidelines on routine use of NGS for clinical practice, as for any other technology, require an external quality assessment. In early 2017, Sciensano’s Cancer Center & Dpt. Quality of laboratories launched a benchmarking trial. The benchmarking trial characteristics are summarized in Table 2. Technical assays were done for solid and hematological tumors, BRCA genes specifically, and on variant interpretation for solid and hematological tumors. There were 16 participating laboratories in 2017 (1 benchmark on solid tumors), 15 laboratories in 2017 (2 benchmarks in hematological tumors) and 12 laboratories in 2018 (1 benchmark on BRCA). In 2020, two benchmarks have been organized (1 in March and another in September 2020). Individual assessments were sent back to participating laboratories, and aggregated results were compiled in a synthesis report available from Sciensano’s Quality of Laboratories Dpt. website. Whereas the majority of participating laboratories were correctly identifying the targeted variants, variant interpretation, and clinical reporting were more of an issue [12]. Therefore, in 2019 national guidelines were published specifically to clarify how to interpret and report variants. There was a close interaction between the coordination at Sciensano and the Bioinformatics unit to assist in the identification of false negatives/positives.

The objectives of Action 4/5 were to assess the quality of NGS tests carried out by the participating laboratories including: the ability of the validated/accredited NGS assays to accurately and reproducibly detect mutations. Lessons learned from the organization of these benchmarks

between 2017 to 2021 which cover different aspects of NGS testing and pathologies, have highlighted areas of improvement for laboratories and will serve as a baseline for the establishment of a sustainable national EQA. Quality frameworks also require meaningful operations reports for continuous evaluation and feedback. The Healthdata.be platform contributes substantially to the infrastructural reporting system in Belgium (<https://healthdata.sciensano.be/fr/home>). This web platform federates health information from multiple national health registers. The platform was created within the Belgian e-Health 2013–2018 Action Plan, and was chosen as tool to securely centralize NGS data generated by licensed and accredited labs (Action 6). The first NGS test was successfully registered on July 1, 2019. The molecular variants are now registered and processed routinely using the ‘Variant Calling File’ format and a secured file transfer protocol. The VCF was a starting point for creating a national data base. The mutations in the genes linked to reimbursed targeted therapies are stored in the Pitter database (which is hosted on Healthdata). Plans are under development in the revision of the NGS technical guidelines to add fusions and CNV annotations for RNA sequencing.

Yet, some challenges remain for quality improvements in NGS use and performance. In terms of data sharing, the ComPerMed experts would like for a core set of NGS metrics to be reported by clinically relevant subgroups (sex, age) (71 %), and reported internationally (64 %) ; whereas reporting NGS metrics by geographic areas was deemed less useful for decision-making. The clinicians wish that this platform is used to drive patient-matching tools wherein clinicians could fine-tune their diagnosis through investigating prior outcomes with treatments of patients with similar molecular profiles in the tumors (i.e. matching). Moreover, the NGS data registered on Healthdata are not augmented with clinical information and/or outcomes and this is a limitation. During the pilot study (Action 9), the beneficiaries commented on issues with the registration of standardized and interpreted patient reports, arguing that this point is crucial for improving the quality and use of NGS data for decision making in policy and practice. The clinician’s report cannot always be linked to the initial NGS test result which hampers the possibility of studying more long term outcomes. Information from the VCF file are also not transferred to the Belgian Cancer Registry (only the actionable mutations) which limits the scope for governance, quality assurance, training, and data reporting, surveillance and monitoring. All indicated, the need for a standardized format for the VCF file in order to allow for comparison and integration of NGS data. A more detailed report is being completed on these particular aspects. A clinical trial is underway to evaluate what is needed to capture follow up of NGS testing: “The Belgian Molecular Profiling Program of Metastatic Cancer for Clinical Decision and Treatment Assignment”.

2.2. Ensuring the dissemination of the NGS Quality framework

With respect to dissemination of the NGS Roadbook, the ComPerMed management Board indicated that they were well aware of the aims and objectives of the intervention, and that the Roadbook had circulated appropriately among the institutions (Supplementary Table 2).

Table 2
General overview of benchmarks characteristics.

Benchmark	2017/1	2017/2	2018/1	2019/1	2019/2	2020/1	2020/2
Targeted cancer types	Colorectal carcinoma, pulmonary carcinoma	Acute myeloblastic leukemia, myeloproliferative neoplasia and (pre-fibrotic) primitive myelofibrosis, myelodysplastic syndromes	BRCA1/BRCA2 genes	Colorectal carcinoma, pulmonary carcinoma, melanoma, gastrointestinal stromal tumor	Acute myeloblastic leukemia, myeloproliferative neoplasia and (pre-fibrotic) primitive myelofibrosis, myelodysplastic syndromes	Colorectal carcinoma, melanoma, gastrointestinal stromal tumor	Acute myeloblastic leukemia, myeloproliferative neoplasia and (pre-fibrotic) primitive myelofibrosis, myelodysplastic syndromes
Sample origin	Mix of engineered and pure cell lines	Mix of engineered and pure cell lines	Mix of pure cell lines	Theoretical case studies	Theoretical case studies	Extracted human genomic DNA from FFPE tumors and FFPE slices from tumors	Mixture of human genomic DNA from the reference cell line GM24385 and synthetic DNA constructs
# participants	16	16	12	20	23	19	20
Variants considered for evaluation	Variants reported by at least two-thirds of participants and validated by exome sequencing and/or ddPCR	Variants reported by at least two-thirds of participants and validated by ddPCR	Variants reported by at least two-thirds of participants and validated by exome sequencing and/or ddPCR	Variants with expert consensus for biological and/or clinical interpretation	Variants with expert consensus for biological and/or clinical interpretation	Variants reported by at least two-thirds of participants	Variants reported by at least two-thirds of participants and validated by ddPCR
# samples	4	3	3	8	6	3	3
# total reported variants	16	45	5	11	25	8	?
# evaluative variants	16	16	6	11	23	3	5

Abbreviations: ddPCR (digital droplet polymerase chain reaction).

NOTES: 1) In the first three benchmarks there were many more variants that were included to cover the genes in the convention. The experts asked for less variants, to explore clinical aspects related to interpretation. In 2020, to be as close to “real-world” conditions as possible, two benchmarks were organized with real tumor samples or with artificial samples mimicking real tumor samples with only few variants as which is observed in clinical practice.

NOTES: 2) More detailed information on the implementation assessments for the NGS Roadbook actions is provided in Supplementary material.

Outreach and consultations with (hemato-)oncologists and laboratories was conducted via face-to-face meetings, the ComPerMed website (<https://www.ComPerMed.be/presentation-ComPerMed>), and the mailing list. They largely agreed that the recommendations in the Roadbook were actionable and that these facilitated coordinated action and a way forward for stakeholders (i.e. with the development of hospital networks in compliance with Action 10).

The NGS Roadbook also benefited from the establishment of a network-infrastructure between hospitals and laboratories in Belgium (Action 10) which provided a mutual learning platform to further: [1] share and disseminate best practices in genomic medicine [2]; facilitate the coordination and expansion of existing genomic medicine implementation efforts [3]; develop new collaborative projects in diverse settings and populations [4]; integrate expertise in complex diagnostics [5], mutualize expenses and [6] harmonize NGS testing and patient care in Belgium. For now, ten networks have been officially approved, that include 115 hospitals and 30 laboratories; the list is available from the RIZIV/INAMI Health Insurance Agency website (<https://www.inami.fgov.be>). High performance NGS warrants investments in infrastructure and technology, trained laboratory staff and in IT which can be mutualized in a network structure. The NGS network establishment is part of the general reform of the hospital care landscape in Belgium [13].

Nonetheless, there were many issues to address including working out the technical, legal and organizational aspects of the Service Level Agreement (SLA) which had not been predefined. The Molecular Advisory Boards were set up independently in each of the networks; and this is another item which could call for further guidance and harmonization from the ComPerMed in the future. The Molecular Advisory Board members discuss the results of the NGS tests in each of the networks, and there were different practices to estimate the number of patients eligible for an NGS test via the Molecular Oncology Consult - a board in which tests and therapies are recommended. Differences in the attitude of laboratories versus hospital management were also observed in relation to the SLA.

2.3. Stakeholder engagement and capacity-building

Optimal use of NGS demands keeping stakeholders engaged to adopt and maintain the new practices. The NGS Roadbook intervention benefited from a transition period where the introduction of NGS testing itself was monitored in the field. As planned in the Roadbook, a national pilot study (Action 9) was launched through an endorsed NGS Convention by the RIZIV/INAMI. The pilot study (2019–2022) aimed to monitor the uptake of NGS testing (guidelines), reporting, and reimbursement in routine (hemato-)oncology. To be reimbursed for testing, hospitals and laboratories with expertise in medical oncology, hematology, anatomic pathology, clinical biology and genomic analysis had to organize themselves within networks (Action 10), participate in the benchmarking trial, provide their legal and technical specifications (Action 4/5) and register NGS data (Action 6).

The NGS Convention has allowed to: i) introduce NGS in clinical routine diagnostics in a controlled manner; ii) guarantee access to this new technology to the healthcare workforce with planned trainings (i.e. in Action 7, ComPerMed's Molecular Pathology Working Group designed a 2-day course which is now mandatory for all residents in pathology); iii) ensure uniform and optimal execution of NGS from the moment of sample collection until the final report. An extension of the convention is under discussion with the Minister of Health.

Overall, there were some difficulties however with upholding the reimbursement of all performed NGS tests as the initial budget was underestimated on the one hand (2 Million € allocated for reimbursement in the NGS Roadbook, 400€ per test, and 5000 tests planned per year) and a high participation rate in the pilot on the otherhand (more than 12 000 tests/year revised estimation for future budgeting plans). As part of the EQA (Actions 4/5), laboratories indicated that more budget is

needed in the future to test more markers and/or enlarge the panels, cover increases in the number of tests and increasing cancer incidence, notably for colorectal cancer and lung cancer. Also the initial NGS cost estimates from the KCE feasibility study did not cover: costs for interpreting DNA alterations other than the actionable mutations, costs for validating the assay and per sample, participation in EQAs or overhead costs. Therefore, the next step consists in reevaluating how NGS can be positioned in the current reimbursement system, taking into account clinical utility, economies of scale, and therapeutic and societal needs.

Most of the experts agreed that the NGS Roadbook has been used for decision making and the development of a new regulatory framework for NGS technology in Belgium. They recognized that the Actions of the NGS Roadbook are well integrated in (hemato-)oncology care and service delivery which has led to improvements in NGS testing capacity in Belgium (federal and regional level). The beneficiaries noted however that the NGS Roadbook could be even better aligned with the expectations of policy makers, care and health insurance providers or patients. This can be explained by the fact that the NGS Roadbook is an initial step towards routine implementation of NGS technology and there are differences in the views of the care provider and the policy maker for instance, or differences in values and beliefs of the care provider and the patient. The challenge for the NGS Roadbook is therefore to resolve these differences under one operating framework. Implementation of the NGS Roadbook is still ongoing which leaves room for improvement still, for instance in regards to optimizing resource allocation in healthcare facilities/laboratories for NGS or collection and reporting of NGS metrics.

2.4. Knowledge integration in civil society

Lastly, taking into account patients' values is a powerful tool for increasing the acceptability and sustainability of a healthcare intervention. To this end, Sciensano organized focus groups with cancer patients to assess their informational needs and learn from their experiences (Action 8). Concise recommendations for good informed consent practices were collected and reported to the ComPerMed board at the annual meeting for molecular diagnostics in Belgium. This feedback allowed for constructive exchange on the attitudes, doubts, and expectations of cancer patients to whom NGS testing is offered. Patients touched upon many topics: data governance, secondary use of NGS data, data access and privacy issues, uncertainties and lack of knowledge about precision medicine, etc. To broaden the feedback, the decision was made to organize a citizen forum on the ethical, legal and societal issues in genomics, as well as a large scale online DNA debate. Citizen labs will be launched in the upcoming months for continuous feedback. The ComPerMed management Board recognized that more could be done to increase citizens' literacy and understanding of NGS technology and its clinical, legal and ethical implications. One of the respondents noted the survey with citizens focused somewhat more on a broad understanding of genomic analyses for inherited disorders vs. NGS testing in oncology specifically.

3. Conclusions and future perspectives

The aim of this study was to provide an overview of the implementation of the NGS Roadbook consisting in 10 Actions. This implementation assessment of the Belgian NGS Roadbook contributes to broader goals of better health system performance, and optimal utilization of NGS in (hemato-)oncology in Belgium. The NGS Roadbook Actions were inter-connected. For instance, the revision of the NGS Convention (Action 9 – testing guidelines and reimbursements) had implications for the EQA benchmarks (Action 4/5). The ComPerMed (Action 1) which provides advice on NGS guidelines directly impacts aforementioned Action 9 and 4/5. Healthdata (Action 6) further coordinates the storage of the data generated during the EQA (Quality of laboratories Dpt.).

Table 3

Recommendations based on the NGS Roadbook evaluation.

Action	Description	Recommendations
1	Establish a commission: Commission Personalized Medicine (ComPerMed)	ORGANIZATIONAL: Ensure sustainability for the expert group e.g. by officialization CONTEXTUAL: Participation of the ComPerMed Management board in NGS/genomics international (research) initiative(s). TECHNICAL: Provide guidelines on the establishment of Molecular Advisory Boards across NGS networks ORGANIZATIONAL/TECHNICAL: Upcoming assessments in 2020 including a full technical and variant interpretation assessment for solid and hematological tumors.
2	Development of (variant interpretation) guidelines for NGS use in (hemato-)oncology	TECHNICAL: Setup a gene-variant catalogue for the tumors identified in the NGS convention (see Action 9). Extension of the catalogue to include variants for all genes analyzed in Belgium as part of routine clinical diagnostics and care. Develop application to automatically link the catalogue with the NGS test results TECHNICAL: Integrate NGS testing in the standard reimbursement nomenclature of RIZIV/INAMI (termination of the NGS convention).
3	Development of criteria for NGS use in (hemato-)oncology	ORGANIZATIONAL: Increase coverage and provide guidelines for comprehensive genomic profiling. CONTEXTUAL: Continuous monitoring of the use of molecular tests in routine clinical practice in Belgium.
4&5	Develop and organize a benchmarking trial and EQA for NGS use in (hemato-)oncology	TECHNICAL/ORGANIZATIONAL: Implement a new wave of NGS EQA trials in line also with developments in the RIZIV/INAMI nomenclature – envisage international collaborations and/or integration in EC activities (e.g. by the DG Joint Research Center) TECHNICAL: Enrich the catalogue with clinical report data to have a more standardized NGS data collection that can be used for future research.
6	Implement NGS registration, storage and data management	Link NGS registration to PITTER (The predictive Tests For A Therapeutic Response registry).
7	Provide NGS education and training	TECHNICAL: Develop further specific trainings targeting care providers on the interpretation of tests CONTEXTUAL: Engage a larger public on ELSI regarding the use of genomic information in society (DNA Debate)
8	Informed consent, legal and ethical implications of NGS use in (hemato-)oncology molecular diagnostics	Increase public knowledge on ELSI in genomics Incorporate citizen's input to policy ORGANIZATIONAL: Establishment of a routine reimbursement procedure for NGS testing via the nomenclature (post NGS Convention termination). RIZIV/INAMI Budget revision.
9	Pilot study 'NGS use in routine diagnostics'	TECHNICAL: Implementation of comprehensive genomic profiling.
10	Build on hospital networks for NGS use in (hemato-)oncology	ORGANIZATIONAL/TECHNICAL: Evaluation of the functioning and performance of NGS networks.

NOTES: More detailed information on the implementation assessments for the NGS Roadbook Actions is provided in Supplementary material.

Overall, there was a good integration of the actions included in the NGS Roadbook which was facilitated by having a centralized coordination by the Belgium Cancer Center. By mandate, the Cancer Center routinely interacts with the main cancer stakeholders in Belgium at federal and regional level and the main operators of the NGS Roadbook actions. An essential element of this study is the use of the Health Information (HI)-Impact framework which allowed us to evaluate the implementation of the NGS Roadbook in a more systematic way. By combining the perspectives of the stakeholders and beneficiaries of the intervention, we were also able to gain a more complete understanding of the impact of the NGS Roadbook in practice.

Our findings provide feedback on what worked and what did not work in deploying the intervention. Some of the difficulties we have noted can be used to inform the scaling up or down of the intervention, as well as provide the evidence base for change and corrective actions. For example, ComPerMed respondents picked up on the following: it is important to correlate cancer NGS metrics with diagnosis, prognosis and treatment, and this is not yet possible. Also, the NGS budget should be better allocated in order to cover the reimbursement of new actionable NGS tests, such as RNASeq as well as NGS data registration, handling and exchange. They noted that this will require a continuous conversation between the different stakeholders (RIZIV/INAMI, Cancer Center, Healthdata, Belgian Cancer Registry, Industry).

This study has enabled the identification of areas for further developments. These are listed in Table 3 below and categorized according to organizational, technical, and contextual requirements. To refine our implementation assessment, investigating associations of NGS use with patient health outcomes in Belgium, and the budgetary impact are explicit points we will consider separately in a future study. Very few studies report on the cost-effectiveness on the application of NGS in cancer care, and this is an area where more research is needed [14,15].

Lessons from the implementation of the NGS Roadbook in Belgium could be useful to inform mutual learning exercises at EU level as well. This is the aim of the current innovative Partnership for Action Against

Cancer (iPAAC) gathering 44 partners from 24 European countries. The multidisciplinary consortium Cancer Prevention Europe (CPE) created in 2018 is also exploring ways to scale up successful interventions within countries, and promote their uptake across borders. The European Commission's "Beating Cancer Plan" launched in 2020 further aims to reduce the cancer burden for patients, their families and health systems while envisioning new partnerships with civil society and across sectors.

Broad-based NGS testing can lead to more actionable results and better utilization of targeted therapies in oncology [16,17]. Whereas national NGS guidelines and quality assurance frameworks have been established in several countries such as the US [18–20], Canada [21], Germany [22], or Japan [23,24], a recent European study also underlined remaining challenges associated with personalized oncology in general [25]. Across countries, issues relate to effective evidence generation, data storage, regulations, value determination, reimbursement and training [3,7,19,22,23,25,26]. Precision oncology also raises the stakes for countries to use the full potential of newer types of data and data systems. For example, Artificial Intelligence may prove useful for NGS data interpretation [27], nonetheless, this requires further dialogue on the strategies for national and EU NGS data collection, access, and sharing of evidence on molecular tests, treatments and patient outcomes.

Belgium's multi-action Roadbook is an original endeavor to facilitate the uptake of NGS testing in clinical practice. Yet, might the Belgian NGS Roadbook approach also work elsewhere? To increase the impact of an intervention and transfer it from one setting to another, further adaptations are required based on the broader clinical, social and political context [9,28]. When it comes to systemic change and the adoption of innovations in the community setting [29], stakeholders inevitably negotiate interests and positions. In precision oncology, the perspectives of the patient and their providers [30], academics [31], health policy planners, industry, regulatory bodies and insurance commissioners are crucial to reconcile.

Funding

This work was supported by the Marie Skłodowska-Curie Actions Individual Fellowship [grant agreement No 795051]; the EU's Horizon 2020 research and innovation programme [OncoNGS grant agreement No 874467]; and the National Institute for Health and Disability Insurance in Belgium (RIZIV/INAMI). The sponsors had no involvement in this study. The Innovative Partnership for Action Against Cancer which is co-funded by the European Union's Third Health Programme (grant agreement N°801520).

Acknowledgements

All authors made substantial contributions to the following: (1) the conception and design of the study (MD, MvdB), the acquisition of data (MD, AH, AA, EVV, WVF), and to the analysis and interpretation of data (MD, MvdB, AH, AA, EVV, AW, WVF). MD drafted the article and all co-authors read and revised it critically. All authors have approved the final submitted version of the manuscript.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.semcancer.2021.06.010>.

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